

Group EE17 Project Report on AI-powered intelligent software application for image-based pattern recognition on Healthy Tomato vs Tomato early blight

The Tomato Disease Classifier was developed using the PlantVillage dataset obtained from Kaggle. The dataset contains labeled images of healthy tomato leaves and tomato leaves affected by Early Blight, making it suitable for supervised image classification using deep learning. The application allows users to upload a tomato leaf image through a Streamlit web interface, after which the trained Convolutional Neural Network (CNN) predicts whether the leaf is healthy or infected and displays the prediction confidence. During development, challenges included configuring the Python virtual environment, GPU compatibility, model path management, and package dependency issues. Deployment on Streamlit Community Cloud was further complicated by TensorFlow's incompatibility with the default Python 3.14 environment. Future improvements include supporting more tomato diseases, enhancing the user interface, improving model accuracy with additional training data, and deploying on a platform with full TensorFlow compatibility.

Group members:

Name	Registration Number	Github Username
John Nkpoikanke Monday	22/EG/EE/2108	whizmonero
Effiong, Elisha Emmanuel	22/EG/EE/2008	ellisselma
Obot, Mmokutmfon Daniel	22/EG/EE/2018	mmokutmfonobot
Abiodun, Godswill Alexander	22/EG/EE/2068	godswillalexander
Udo Aniekpeno Nkereuwem	22/EG/EE/1968	anexnk
Jackson Hannah Udoh	22/EG/EE/2078	darannah
Emmanuel Donald Uwem	23/EG/EE/008	DrDonney
Wobo, Prosper Nyeyiruchi	22/EG/EE/1978	Pan-da10
John, Wisdom Akan	22/EG/EE/2098	Don-Wizzy-088
Archibong, Oyokunyi Uwe	22/EG/EE/1988	Sammye53
Praise Nkereuwem Esikhene	22/EG/EE/2038	praiseesikhene81-cyber
Gideon Akuche ifeanyichukwu	23/EG/EE/098	bigdheon